

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE423: Computer Graphics

Assignment 04 | Total Marks: 20 | Summer 2025

Deadline: September 12, 2025

Q1.	a.	How do the HSL and HSV color models fundamentally differ in their representation of color attributes, and how are these differences reflected in their use within digital imaging and user interface design? [4]
	b.	In a packaging design project for a beverage brand, a creative designer selects a vibrant color using the HSV model: Hue (H) = 45° , Saturation (S) = 0.8, and Value (V) = 0.9. Since the final design will be printed using CMY-based ink, the HSV values must be converted into the CMY (Cyan, Magenta, Yellow) color model. Compute the corresponding CMY values and illustrate the approximate position of this color within the CMY color model. [6]

Q2.	a.	In a 3D simulation of an ancient temple, two light sources illuminate a ceremonial chamber. One light is at (5, 10, 7) with intensity 20 and radius 25, and the other is at (15, 30, 45) with intensity 10 and radius 60. You are asked to calculate the illumination at point (0, 2, 4) on the xz-plane, where a magical symbol lies. The camera is positioned at (3, 2, 6). The surface has ambient, diffuse, and specular coefficients of 0.2, 0.6, and 0.4, respectively, and a shininess factor of 10. The ambient light intensity is 8. If the ambient light intensity is 8 and the shininess factor is 10, calculate the total reflected light intensity at the point using Phong's Lighting Model with the attenuation factors. [6]
	b.	In a 3D animation project using Blender, we are applying color to a quadrilateral ground tile. The quadrilateral is defined by the vertices A(-200, 250), B(200, 250), C(200, 150), and D(-200, 150). Let point P be the centroid of the quadrilateral. The color at vertex A is specified in the RGB model with values (0.7, 0.2, 0.5). The color at vertex C is specified in the CMY model with values (0.2, 0.3, 0.4). Using Gouraud shading, compute the interpolated color at point P and express the result in the RGB color model. [4]